Cyber-Physical Attack Detection in Smart Grids Using Graph Signal Processing

Aniket Rajak
Department of Information Technology
Indian Institute of Information Technology
Sonepat, India
aniketrajak849@gmail.com

Rushil Jain
Department of Information Technology
Indian Institute of Information Technology
Sonepat, India
rushiljain69@gmail.com

***Abstract*—False data injection attacks (FDIA) are a major cyber-physical threat to smart grids because they can manipulate control decisions while bypassing conventional residual-based bad data detection. This report presents an empirical and theoretical comparison of three detection methods: residual bad data detection (BDD), nodewise statistical Z-score detection, and graph signal processing (GSP)-based high-frequency anomaly detection. A reproducible Python simulation was implemented on a connected 30-bus graph with a linear measurement model and two attack classes: random FDIA and stealth FDIA designed in the measurement model subspace. Experimental results show that residual BDD detects random attacks well but fails on stealth attacks, while Z-score detection underperforms in both scenarios. GSP achieves the best overall attack detection with strong stealth-attack recall and low false positive rate. The findings support the use of topology-aware graph spectral methods for smart-grid cyber-physical anomaly detection.**

Keywords— Smart Grid, False Data Injection Attack, Graph Signal Processing, Cyber-Physical Security, Anomaly Detection

I. INTRODUCTION

Smart grids combine electrical infrastructure with communication and computation layers, enabling real-time monitoring and automated control. This cyber-physical coupling improves efficiency but increases attack surface. Among known threats, FDIA is particularly dangerous because the attacker can manipulate measurements and influence state estimation and downstream decisions.

A key challenge is that many practical monitoring pipelines rely on residual checks from state estimation. Prior literature shows that coordinated attacks can be crafted to evade these checks. This motivates structure-aware approaches that exploit physical network topology in addition to raw measurement consistency.

This report evaluates whether GSP can improve FDIA detection, especially for stealth attacks that evade classical residual-based checks.

II. BACKGROUND AND CONTEXT

Smart grids integrate power systems with communication networks, improving efficiency but increasing exposure to cyber-physical attacks. Among these, **False Data Injection Attacks (FDIA)** are critical, as they can manipulate measurements while bypassing traditional residual-based detection.

Conventional methods such as **Bad Data Detection (BDD)** rely on measurement inconsistency and fail against coordinated (stealth) attacks. While machine learning methods improve detection, they require labelled data and may lack generalization.

Graph Signal Processing (GSP) provides a topology-aware framework by analysing signals over the grid structure. This enables detection of anomalies that appear as irregularities in the graph spectral domain.

III. RELATED WORK

The foundational FDIA vulnerability is established in the smart-grid state-estimation context by Liu, Ning, and Reiter, who show attack vectors that can bypass traditional bad-data detection while causing arbitrary state errors.

Classical state estimation and bad-data detection formulations are detailed in standard power-system estimation literature and comparative BDD studies. Later work introduced sparse-optimization and machine-learning based detectors, and several deep-learning approaches improved detection rates under certain data assumptions.

In parallel, GSP developed a formal framework for signals defined on irregular domains (graphs), including graph Fourier analysis and graph spectral filtering. Recent power-system work has started applying GSP concepts for bad-data/anomaly detection.

IV. SYSTEM AND THREAT MODEL

A. *Measurement Model*

The simulation uses a linearized measurement model:

$$\boldsymbol{z} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{e}$$

where z is the measurement vector, H is the measurement matrix, x is the state vector (bus-level state signal), and e is Gaussian noise.

B. *Threat Model*

The attacker can manipulate measurement data and has knowledge of the system model H. The goal is to alter state estimation while remaining undetected. The system assumes Gaussian noise and fixed grid topology.

C. *Attack Model*

Two FDIA classes were used:

- Random Attack: sparse, high-magnitude perturbations on measurements.
- Stealth Attack: model-consistent attack with a = Hc, giving:

$$\boldsymbol{z}' = \boldsymbol{z} + \boldsymbol{H}\boldsymbol{c}$$

For this attack type, residual-based detection is theoretically weak because the attack lies in the measurement model subspace.

D. *Graph Model and GSP Score*

The grid is represented as a graph G = (V, E) with Laplacian $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{A}$. The estimated state signal $\hat{\boldsymbol{x}}$ is projected onto graph spectral components. The detector score is defined as the high-frequency energy ratio:

$$\boldsymbol{score_{gsp}} = \frac{\left\|\boldsymbol{U_{HF}^T}\hat{\boldsymbol{x}}\right\|_2^2}{\|\boldsymbol{U^T}\hat{\boldsymbol{x}}\|_2^2}$$

where U is the Laplacian eigenvector matrix and $\boldsymbol{U_{HF}}$ selects high-frequency eigenmodes.

V. Compared Detection methods

A. *Residual BDD*

- Residual score:

$$\boldsymbol{score_{res}} = \frac{\|\boldsymbol{z} - \boldsymbol{H}\hat{\boldsymbol{x}}\|_2}{\sqrt{\boldsymbol{m}}}$$

- Decision: attack if $\boldsymbol{score_{res}}$ exceeds a normal-data threshold.

B. *Nodewise Z-score Baseline*

- Using training normal data mean mu and std sigma, the score is:

$$\boldsymbol{score_z} = \boldsymbol{max_i}\left|\frac{\hat{\boldsymbol{x}}_{i-}\boldsymbol{\mu_i}}{\boldsymbol{\sigma_i}}\right|$$

- Decision: attack if $\boldsymbol{score_z}$ exceeds a normal-data threshold.

C. *GSP high-frequency Detector (proposed)*

- Decision: attack if $\boldsymbol{score_{gsp}}$ exceeds a normal-data threshold.

VI. Setup and implementation

A. *Workflow Overview*

The proposed detection pipeline follows a structured sequence:

1) Generate a connected graph representing the power grid topology.
2) Construct the graph Laplacian matrix $\boldsymbol{L} = \boldsymbol{D} - \boldsymbol{A}$.
3) Generate smooth graph signals representing normal system states.
4) Apply measurement model $\boldsymbol{z} = \boldsymbol{Hx} + \boldsymbol{e}$.
5) Inject attack vectors (random or stealth).
6) Estimate system state using least squares:

$$\hat{\boldsymbol{x}} = (\boldsymbol{H^T H})^{-1}\boldsymbol{H^T z}$$

7) Compute detection scores:
   - Residual BDD
   - Nodewise Z-score
   - GSP high-frequency energy ratio
8) Apply threshold-based classification.

B. *Implementation Notes*

The system is implemented in Python using:

- **networkx** for graph generation
- **numpy** and **scipy** for numerical computation
- matplotlib for visualization

A connected random geometric graph with 30 buses and 60 measurements is generated. The measurement matrix H is randomly initialized and ensured to be full rank. Graph spectral decomposition is performed using eigenvalue decomposition of the Laplacian.

High-frequency components are identified by selecting the upper portion of eigenmodes, and the GSP score is computed as the ratio of high-frequency energy to total signal energy.

Thresholds are determined using the 99.5th percentile of validation data (aux_stats.json) to maintain low false positive rates.

C. *Experimental Setup*

- Graph: connected 30-bus random geometric graph.
- Measurements: 60 channels.
- Noise: Gaussian, std = 0.05.
- Data splits:
  a. Train normal: 1400
  b. Validation normal: 400
  c. Test normal: 800
  d. Test random attack: 500
  e. Test stealth attack: 500
- Thresholding: 99.5th percentile of validation-normal score for each detector.

Outputs were generated by the script at run_fdia_gsp_experiment.py

VII. Results

A. *Practical Comparison (measured performance)*

Method	*Scenario*	*Recall*	*FPR*	*F1*	*AUC*
Residual BDD	Overall	0.504	0.0137	0.665	0.749
Residual BDD	Random Attack	1.000	0.0137	0.989	1.000
Residual BDD	Stealth Attack	0.008	0.0137	0.016	0.498
Z-score Nodewise	Overall	0.002	0.0087	0.004	0.509
Z-score Nodewise	Random Attack	0.004	0.0087	0.008	0.511
Z-score Nodewise	Stealth Attack	0.000	0.0087	0.000	0.506
GSP High-Frequency	Overall	0.880	0.0025	0.935	0.989
GSP High-Frequency	Random Attack	0.810	0.0025	0.893	0.981
GSP High-Frequency	Stealth Attack	0.950	0.0025	0.972	0.997

Performance is evaluated using recall, false positive rate (FPR), F1-score, and AUC.

B. *Key empirical evidence for stealth vulnerability*

Mean residual score by class:

- Normal: 0.03499
- Random attack: 0.70810
- Stealth attack: 0.03492

This directly confirms the stealth behavior: stealth attack residual is nearly indistinguishable from normal, while random attack residual is large.

C. *Figures*

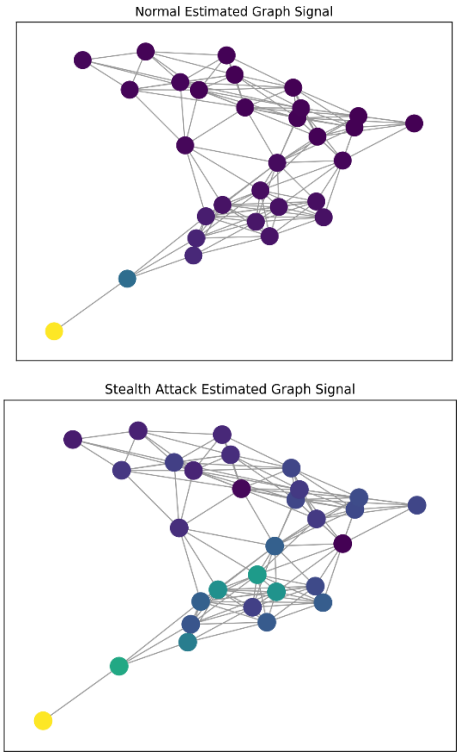

Fig.1. Graph Signal Example

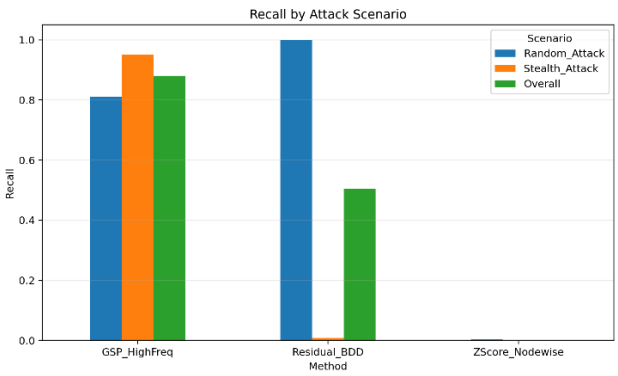

Fig.2. Recall By Scenario

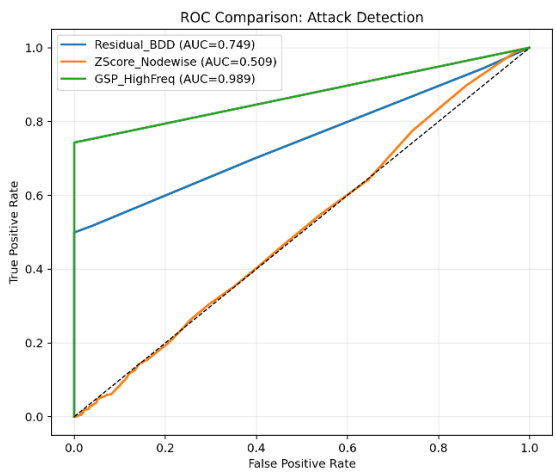

Fig.3. ROC Comparison

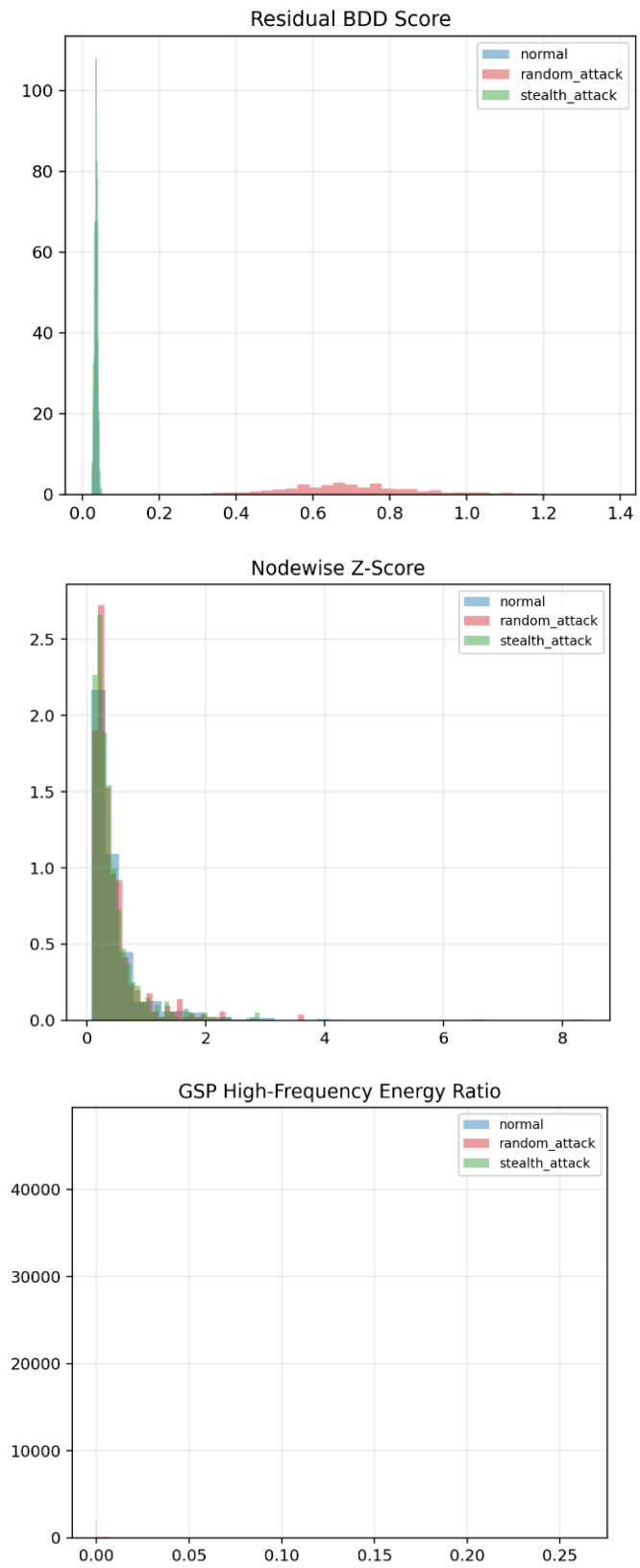

Fig.4. Score Distributions

State estimation is performed using least-squares, and detection scores are computed using residual, Z-score, and GSP-based high-frequency energy metrics.

As shown in Fig. 2 and Fig. 3, GSP significantly outperforms baseline methods, particularly under stealth attack conditions.

VIII. Analysis and Security Discussion

A. *Theoretical explanation*

1) *Residual BDD blind spot:*
   - If attack vector is a = Hc, then $z' = z + Hc$ remains consistent with model structure.
   - The estimator absorbs attack as state shift c, keeping residual nearly unchanged.
2) *Nodewise magnitude blind spot:*
   - Coordinated attacks can keep per-node deviations small and avoid pointwise thresholds.
3) *GSP advantage:*
   - Even small coordinated perturbations can inject graph-high-frequency content (non-smooth behavior across connected buses), which GSP explicitly measures.

B. *Practical evidence from this run*

1) Residual BDD achieved near-perfect random attack recall but almost zero stealth recall.
2) Z-score method remained near-random performance.
3) GSP achieved strong detection in both random and stealth scenarios, with the best AUC and lowest FPR.

C. *Security Discussion*

Residual BDD performs well for random attacks but fails for stealth attacks, as residual values remain similar to normal data. Z-score methods also underperform due to lack of structural awareness.

The GSP method exploits graph topology to detect **high-frequency irregularities**, enabling effective detection of both attack types. This shows that topology-aware methods are more robust against coordinated attacks.

IX. Limitations and Future Work

A. *Limitations*

1) The study uses a synthetic random geometric graph, which may not fully capture real-world grid topology.
2) The measurement model is linear and does not account for nonlinear AC power flow dynamics.
3) The attacker is assumed to have knowledge of the system model, which may not always hold in practice.
4) Detection is based on static snapshots and does not incorporate temporal dynamics.
5) Threshold selection is based on validation data and may require tuning for different systems.

B. *Future Works*

1) Extend the approach to real IEEE benchmark systems (e.g., IEEE 14-bus, 118-bus).
2) Incorporate temporal models such as RNNs or Kalman filtering.
3) Develop adaptive thresholding mechanisms.
4) Combine GSP with machine learning models for hybrid detection.
5) Evaluate robustness under partial observability and missing data.

X. Conclusion

This study demonstrates that topology-aware graph spectral detection can substantially improve FDIA detection in smart grids, especially against stealth attacks that bypass residual-based checks. In this experiment, GSP provided the best tradeoff between recall and false alarms and showed a clear advantage in stealth attack detection.

XI. Reproducibility Notes

Run Script: run_fdia_gsp_experiments.py
Main Outputs:

- results/metrics_summary.csv
- results/test_scores.csv
- results/aux_stats.json

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